DOCKER

Steps:

1)open ubuntu with the ec2 instance created and change the access to root user

2)install docker engine

3)start docker and verify whether the service is running

4)create a docker network wordpress and mysql communication

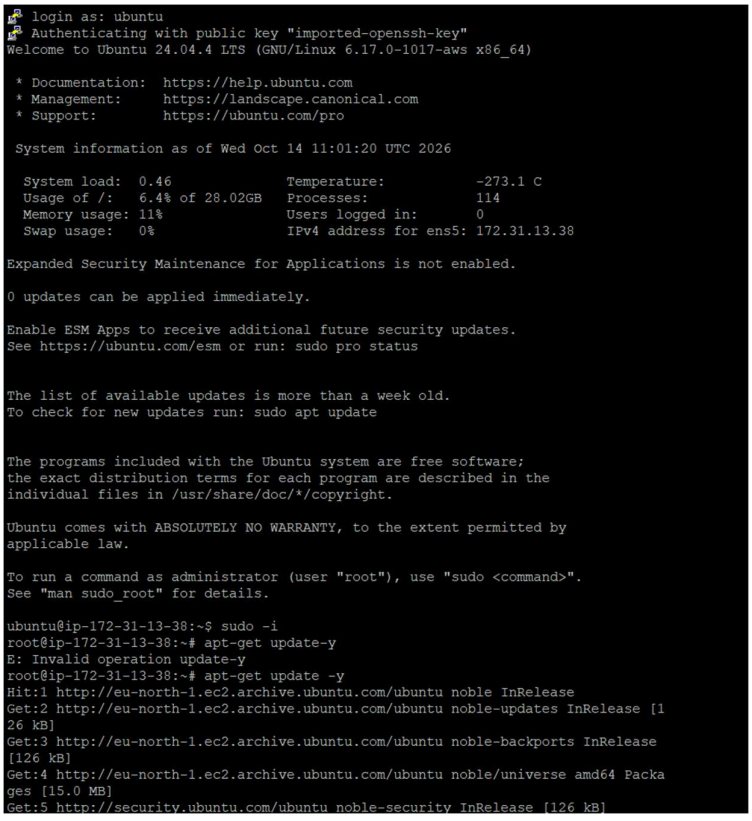
5)create a persistant docker volume and launch mysql container with a database,username and password

6) Download the WordPress image and launch the WordPress container with MySQL connection settings.

7) Configure the required inbound port in the AWS EC2 security group.

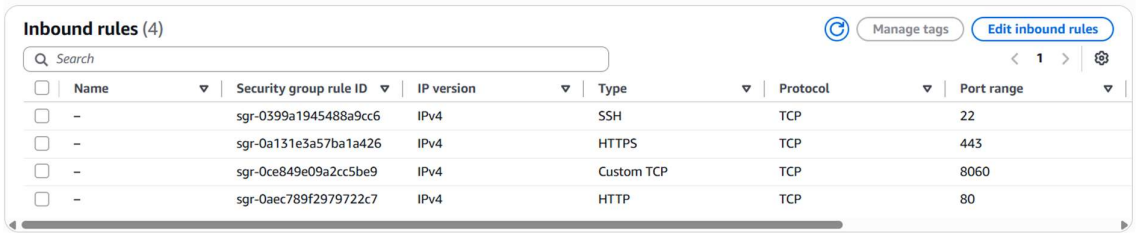
8) Access WordPress using the EC2 public IP and mapped port, complete the installation, and verify database connectivity.

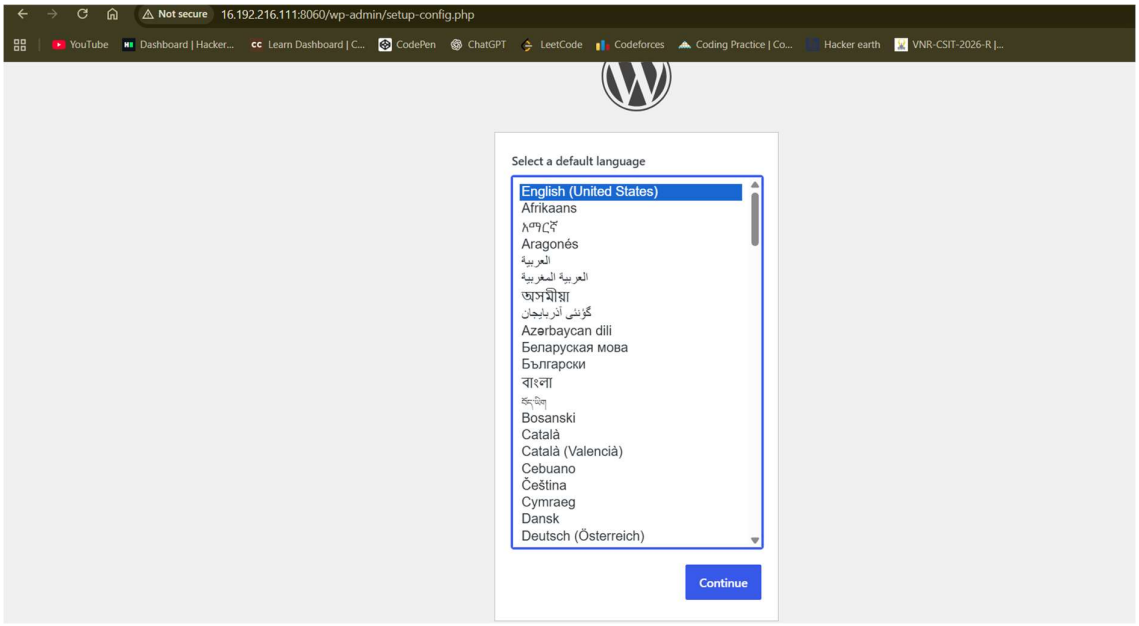
Wordpress:



```
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation
-en [5982 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Compo
nents [3871 kB]
Get:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f
 Metadata [301 kB]
Get:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Pac
kages [269 kB]
Get:10 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translat
ion-en [118 kB]
Get:11 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Co
mponents [35.0 kB]
Get:12 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-
n-f Metadata [8328 B]
Get:13 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
Packages [1397 kB]
Get:14 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Transl
ation-en [308 kB]
Get:15 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
Components [181 kB]
Get:16 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
c-n-f Metadata [17.7 kB]
Get:17 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe am
d64 Packages [1702 kB]
Get:18 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Tr
anslation-en [342 kB]
Get:19 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe am
d64 Components [388 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [114
5 kB]
Get:21 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe am
d64 c-n-f Metadata [34.9 kB]
Get:22 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted
amd64 Packages [1805 kB]
Get:23 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted
Translation-en [411 kB]
Get:24 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse
amd64 Packages [50.7 kB]
Get:25 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse
Translation-en [14.0 kB]
Get:26 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse
amd64 Components [940 B]
Get:27 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse
amd64 c-n-f Metadata [656 B]
Get:28 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd6
4 Packages [40.6 kB]
Get:29 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Tran
slation-en [9172 B]
```

```
    }
]
root@ip-172-31-13-38:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED         STATUS
   PORTS                                  NAMES
78c3afded153   wordpress   "docker-entrypoint.s…"   2 minutes ago   Up 2 minutes
   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   wordpress
root@ip-172-31-13-38:~# docker ps -a
CONTAINER ID   IMAGE       COMMAND                  CREATED         STATUS
   PORTS                                  NAMES
78c3afded153   wordpress   "docker-entrypoint.s…"   2 minutes ago   Up 2 minutes
   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   wordpress
root@ip-172-31-13-38:~# docker stop wordpress
wordpress
root@ip-172-31-13-38:~# docker rm wordpress
wordpress
root@ip-172-31-13-38:~# docker run -d --name wordpress -p 8060:80 wordpress
5f6787982adba57c75d6a47747d6d8b8304d9715a186facf7f8bc8a560256f95
root@ip-172-31-13-38:~# docker ps
CONTAINER ID   IMAGE       COMMAND                  CREATED         STATUS
   PORTS                                  NAMES
5f6787982adb   wordpress   "docker-entrypoint.s…"   8 seconds ago   Up 7 seconds
   0.0.0.0:8060->80/tcp, [::]:8060->80/tcp   wordpress
root@ip-172-31-13-38:~# docker port wordpress
80/tcp -> 0.0.0.0:8060
80/tcp -> [::]:8060
root@ip-172-31-13-38:~# docker logs wordpress
WordPress not found in /var/www/html - copying now...
Complete! WordPress has been successfully copied to /var/www/html
AH00558: apache2: Could not reliably determine the server's fully qualified doma
in name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress t
his message
AH00558: apache2: Could not reliably determine the server's fully qualified doma
in name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress t
his message
[Wed Oct 14 11:11:30.369629 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Ap
ache/2.4.68 (Debian) PHP/8.3.35 configured -- resuming normal operations
[Wed Oct 14 11:11:30.369758 2026] [core:notice] [pid 1:tid 1] AH00094: Command l
ine: 'apache2 -D FOREGROUND'
```





MYSQL

```
root@ip-172-31-13-38:~# docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
514ea420f997: Pull complete
e896da32f635: Pull complete
029a2ef99eb4: Pull complete
a798338f8ea0: Pull complete
d41b77484029: Pull complete
d057682167d5: Pull complete
ecb59e1f4a8f: Pull complete
1a710d5def08: Pull complete
c233331fdce7: Pull complete
d6526c3b0b71: Pull complete
9e6faf632f7d: Download complete
2659fa5447a2: Download complete
Digest: sha256:893418c8a7a34c60d2d961cc30ef136095da141478562ed2101ae3f5faba9c3f
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
```



```
root@ip-172-31-13-38:~# docker image inspect mysql
[
    {
        "Id": "sha256:893418c8a7a34c60d2d961cc30ef136095da141478562ed2101ae3f5faba9c3f",
        "RepoTags": [
            "mysql:latest"
        ],
        "RepoDigests": [
            "mysql@sha256:893418c8a7a34c60d2d961cc30ef136095da141478562ed2101ae3f5faba9c3f"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-10-12T20:11:44.489747546Z",
        "Config": {
            "ExposedPorts": {
                "3306/tcp": {},
                "33060/tcp": {}
            },
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "GOSU_VERSION=1.19",
                "MYSQL_MAJOR=innovation",
                "MYSQL_VERSION=26.7.0-1.el9",
                "MYSQL_SHELL_VERSION=26.7.0-1.el9"
            ],
            "Entrypoint": [
                "docker-entrypoint.sh"
            ],
            "Cmd": [
                "mysqld"
            ],
            "Volumes": {
                "/var/lib/mysql": {}
            },
            "WorkingDir": "/"
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 271641479,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:48e226befb1f0b8f298f83bce36062909a1c72a62b3172ba45371967ce78465f",
                "sha256:206e24b7c10432595cc2a0e803de450b188e8956243929c9957521f4080fa620",
                "sha256:61aac406503a926c59ce72e5fc30192ed340baf9c5132f039806bd91fa3c7424",
                "sha256:feedac4593f1355e5ca8e7be11de9fc4d633dcdeb2ccf20c991ce03ebf19e163",
                "sha256:506f9ea77e5cf956d0a98b68c630dcd6a827c6675cdbab5114b0172eeb80da82",
                "sha256:2575672a86af6e1c9306a31915f48f529e1894e35098b8fe1cd22483a91f27b7",
                "sha256:e3cb3df10965e1741a6e1d6a32309a414f1006e82f485ac644bae835755d2182",
                "sha256:99f25ffc33c7344dc3b024a000d1997e1951ac48ad248d1072478cac4edbc94a",
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-10-14T11:25:20.674102084Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:893418c8a7a34c60d2d961cc30ef136095da141478562ed2101ae3f5faba9c3f",
            "size": 2619
        }
    }
]
```

```
root@ip-172-31-13-38:~# docker run -d \
  --name mysql-db \
  -e MYSQL_ROOT_PASSWORD=my_root_pw \
  mysql:latest
5870958f72faf2f779b37cb244c83344cde1ef9cfa5cb53e5c5bf2a8876c9998
root@ip-172-31-13-38:~# docker exec -it mysql-db mysql -u root -pmy_root_pw -e "SHOW DATABASES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
```

Inbound rules (5)

Search

Manage tags | Edit inbound rules

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Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0399a1945488a9cc6	IPv4	SSH	TCP	22
-	sgr-0a131e3a57ba1a426	IPv4	HTTPS	TCP	443
-	sgr-0c854cc544b41f4d4	IPv4	Custom TCP	TCP	3307
-	sgr-0ce849e09a2cc5be9	IPv4	Custom TCP	TCP	8060
-	sgr-0aec789f2979722c7	IPv4	HTTP	TCP	80